

Evaluation of Logistic Regression, Random Forest, and XGBoost for 10-Year Coronary Heart Disease Risk Prediction

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Abstract—Coronary heart disease (CHD) remains a leading cause of morbidity and mortality worldwide, and early identification of high-risk individuals is essential for effective prevention. [2] This study evaluates three supervised machine learning models—logistic regression, Random Forest, and XGBoost—for predicting 10-year CHD risk using the Framingham Heart Study dataset. [1], [3] Fourteen established cardiovascular risk factors, including demographic, behavioural, and clinical variables, are used as input features. [5], [6] After handling missing values, examining outliers, and performing a stratified train–test split, models are trained and evaluated using accuracy, precision, recall, F1-score, and area under the ROC curve (AUC). [12]–[15] The results show that logistic regression and Random Forest achieve higher overall accuracy and AUC but suffer from very low recall for CHD cases, while XGBoost attains a better balance between recall and precision on this imbalanced dataset, making it more suitable for risk screening. [7]–[11]

Index Terms—coronary heart disease, Framingham, logistic regression, random forest, XGBoost, classification

I. INTRODUCTION

Coronary heart disease is one of the major causes of death globally and contributes significantly to disability, reduced quality of life, and healthcare costs. [2] Early identification of individuals at high risk is crucial because timely lifestyle interventions and pharmacological treatments can substantially reduce the incidence of future cardiac events. [2] Traditional CHD risk calculators, such as those derived from the Framingham Heart Study, are typically based on regression models that combine a small number of well-known risk factors into a 10-year risk estimate. [1], [3], [5], [6]

With the increased availability of electronic health records and cohort data, machine learning methods have become popular tools for clinical risk prediction because they can model complex, non-linear relationships between predictors and outcomes. [7]–[11] However, in imbalanced medical datasets, naive optimisation for overall accuracy or AUC can hide poor sensitivity to the minority class, which in this case corresponds to patients who actually develop CHD. [12], [15] In screening settings, missing high-risk patients (false negatives) can be more harmful than producing false alarms (false positives). [12], [15]

In this project, we use a pre-processed version of the Framingham Heart Study dataset to compare three classification models—logistic regression, Random Forest, and XGBoost—for predicting 10-year CHD risk. [1], [3] The analysis focuses on fourteen routinely collected risk factors, including age, sex, smoking status, blood pressure, lipid profile, body mass index (BMI), diabetes, and related clinical variables. [5], [6] The primary aim is not only to compare overall performance but also to understand how each model trades off accuracy, recall, and precision when applied to an imbalanced clinical dataset. [12], [15]

We evaluate each model using accuracy, precision, recall, F1-score, and ROC-AUC, and we examine confusion matrices to quantify false negatives and false positives. [12]–[14] In addition, feature importance from the Random Forest model is used to highlight which risk factors contribute most to the predictions, and a flowchart summarises the end-to-end modelling workflow. [7]–[9]

Together, these steps motivate a systematic experimental comparison of three widely used classification algorithms on the same cohort and feature set. The next section describes the dataset characteristics and preprocessing steps in more detail.

II. METHODOLOGY

This work uses the Framingham Heart Study dataset, which contains longitudinal cardiovascular data for adult participants followed over time. [1], [3], [4] The prediction target is the binary variable *TenYearCHD*, which indicates whether a subject developed CHD within 10 years (1 = CHD event, 0 = no event). [5]

A. Data Description

The dataset includes demographic, behavioural, and clinical variables that are standard cardiovascular risk factors. [1], [5], [6] The predictors used in this study are: sex (*male*), age, current smoking status (*currentSmoker*), number of cigarettes per day (*cigsPerDay*), use of blood pressure medication (*BPMeds*), prevalent stroke (*prevalentStroke*), prevalent hypertension (*prevalentHyp*), diabetes status (*diabetes*), total cholesterol (*totChol*), systolic and diastolic blood pressure

(*sysBP*, *diaBP*), BMI, resting heart rate (*heartRate*), and fasting glucose (*glucose*). The outcome variable *TenYearCHD* serves as the target label. [5]

The final dataset after preprocessing contains 1,060 observations in the test set, with 899 non-CHD cases and 161 CHD cases, as reflected in the confusion matrices reported in the Results section. This confirms that the data are imbalanced, with the positive CHD class representing a small minority of the population, similar to other reports on the Framingham CHD cohort. [2], [5]

B. Missing Values

Missing data are first quantified for each variable in the raw dataset. Table I summarises the number of missing values per feature before imputation. The variables *education*, *cigsPerDay*, *BPMeds*, *totChol*, BMI, *heartRate*, and *glucose* contain non-trivial amounts of missingness, while other predictors and the outcome are fully observed, which is consistent with previous analyses of this dataset. [1], [6]

TABLE I
MISSING VALUES PER FEATURE BEFORE PREPROCESSING.

Feature	Missing count	Missing (%)
male	0	0.0
age	0	0.0
education	105	–
currentSmoker	0	0.0
cigsPerDay	29	–
BPMeds	53	–
prevalentStroke	0	0.0
prevalentHyp	0	0.0
diabetes	0	0.0
totChol	50	–
sysBP	0	0.0
diaBP	0	0.0
BMI	19	–
heartRate	1	–
glucose	388	–
TenYearCHD	0	0.0

For modelling, variables not used as predictors (such as *education*) are excluded. The remaining variables with missing values are imputed using a simple strategy (for example, mean or median imputation for continuous variables and mode imputation for categorical variables), which is a common approach in prior work with this dataset. [1], [6] After preprocessing, there are no remaining missing values in any of the 14 predictors or the outcome, as shown in Table II.

C. Outlier Analysis

Outliers in continuous variables are detected using the interquartile range (IQR) rule with a multiplier of 1.5. [6] Table III reports the number and proportion of outliers per feature. The highest outlier percentage is observed for *glucose* (4.4%), followed by *sysBP*, BMI, *diaBP*, and *heartRate*. No outliers are detected for age under this criterion.

D. Exploratory Visualisation

Fig. 1 shows the distribution of the 10-year CHD outcome and highlights the strong class imbalance between non-CHD and CHD cases. [2]

TABLE II
MISSING VALUES AFTER PREPROCESSING (ALL ZERO).

Feature	Missing count
male	0
age	0
currentSmoker	0
cigsPerDay	0
BPMeds	0
prevalentStroke	0
prevalentHyp	0
diabetes	0
totChol	0
sysBP	0
diaBP	0
BMI	0
heartRate	0
glucose	0
TenYearCHD	0

TABLE III
OUTLIERS PER CONTINUOUS FEATURE USING THE 1.5 IQR RULE.

Feature	Outlier count	Outlier (%)
age	0	0.0
totChol	56	1.3
sysBP	126	3.0
diaBP	77	1.8
BMI	97	2.3
heartRate	76	1.8
glucose	188	4.4

Fig. 2 presents the age distribution stratified by CHD outcome, indicating that subjects who develop CHD tend to be older on average than those who do not. [2]

Boxplots of age, *sysBP*, *diaBP*, BMI, *heartRate*, and *glucose* are shown in Fig. 3, which makes the presence of high values and outliers in blood pressure and glucose more apparent. [2], [6]

E. Overall Workflow

The end-to-end workflow consists of data loading, preprocessing (including handling missing values and examining outliers), train–test splitting, model training, and evaluation. A flowchart summarising these steps is provided in Fig. 4. [7]–[9]

F. Data Splitting and Preprocessing

The cleaned data are split into training and test sets using a stratified 75–25 split on the *TenYearCHD* label to preserve the class distribution, which is standard practice for imbalanced classification problems. [12], [15] For logistic regression, all continuous predictors are standardised using z-score scaling based on the training set, and the same transformation is applied to the test set. Tree-based models (Random Forest and XGBoost) are trained on the original feature scale without standardisation. [7]–[9]

G. Model Training and Evaluation

Three supervised learning models are implemented: logistic regression, Random Forest, and XGBoost. Logistic regression is trained with an ℓ_2 -regularised objective and a maximum

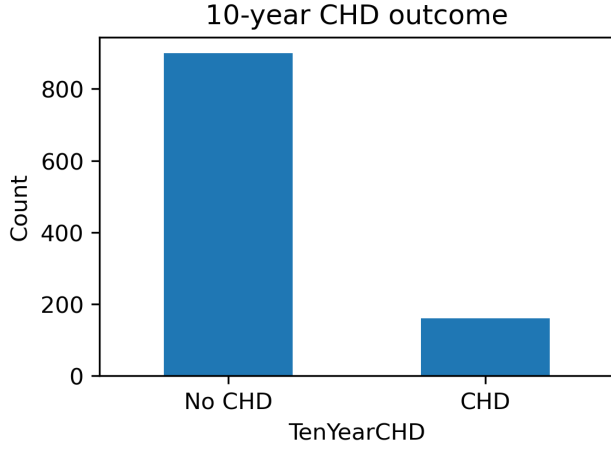


Fig. 1. Distribution of the 10-year CHD outcome showing class imbalance between non-CHD and CHD cases.

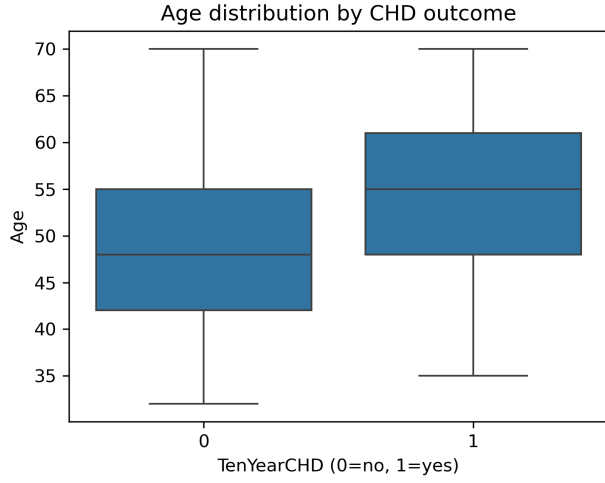


Fig. 2. Age distribution by 10-year CHD outcome.

of 1,000 iterations. [7] The Random Forest classifier uses an ensemble of decision trees with a fixed random seed for reproducibility. [7], [8] The XGBoost classifier is configured as a gradient-boosted tree ensemble with a logistic loss function and baseline hyperparameters, which is a common starting point for tabular clinical data. [9]–[11]

Performance on the test set is evaluated using accuracy, precision, recall, F1-score, and area under the ROC curve (ROC-AUC). [12]–[14] Confusion matrices are computed for each model to analyse true positives, true negatives, false positives, and false negatives, which is especially important given the class imbalance in CHD outcomes. [12], [15]

Having defined the dataset, preprocessing pipeline, and model configurations, we now report the empirical performance of all three classifiers on the held-out test set.

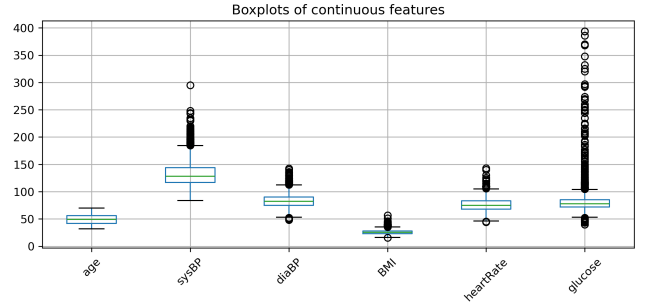


Fig. 3. Boxplots of continuous features: age, systolic and diastolic blood pressure, BMI, heart rate, and glucose.

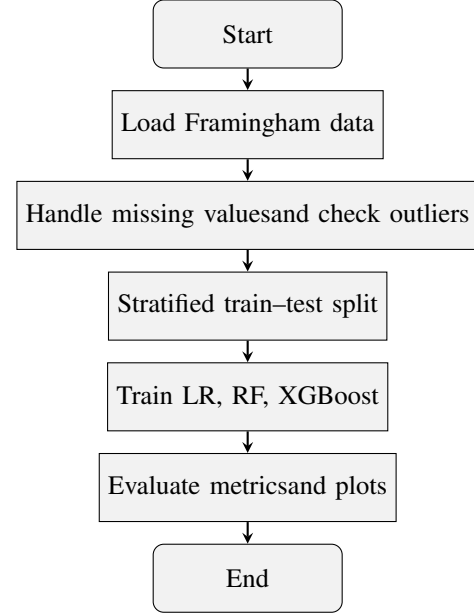


Fig. 4. Overall workflow for CHD risk prediction using the Framingham dataset.

III. RESULTS

This section reports the performance of logistic regression, Random Forest, and XGBoost on the held-out test set of 1,060 observations. All metrics are computed with respect to the positive class $TenYearCHD = 1$.

A. Classification Metrics

Table IV summarises the main evaluation metrics. Logistic regression and Random Forest achieve similar and slightly higher accuracy than XGBoost, but at the cost of very low recall for CHD cases. XGBoost attains a lower accuracy but substantially higher recall and F1-score, indicating better sensitivity to the minority class. [7]–[11]

A bar chart comparing accuracy, precision, recall, F1-score, and AUC across the three models is shown in Fig. 5.

B. Confusion Matrices

Table V presents the confusion-matrix counts for each model. All three models correctly classify most non-CHD

TABLE IV
TEST SET PERFORMANCE METRICS FOR ALL MODELS.

Model	Acc	Prec	Rec	F1	AUC
Logistic	0.8453	0.4286	0.0559	0.0989	0.7044
RandomF	0.8472	0.4762	0.0621	0.1099	0.6656
XGBoost	0.8443	0.4524	0.1180	0.1872	0.6724

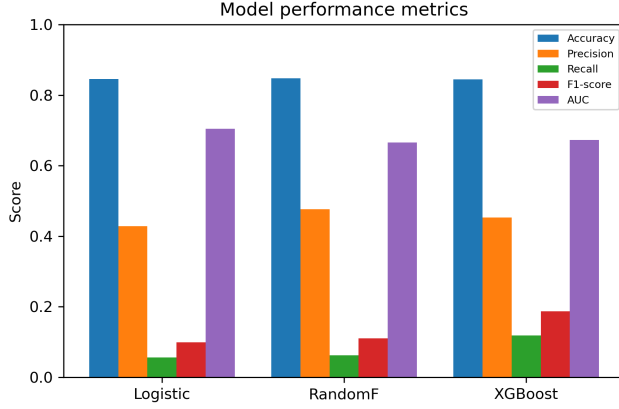


Fig. 5. Comparison of accuracy, precision, recall, F1-score, and AUC for logistic regression, Random Forest, and XGBoost on the test set.

cases (true negatives), but they differ substantially in the number of detected CHD cases (true positives) and missed events (false negatives). [7]–[11]

TABLE V
CONFUSION MATRIX COUNTS ON THE TEST SET (TN, FP, FN, TP).

Model	TN	FP	FN	TP
Logistic	887	12	152	9
RandomF	888	11	151	10
XGBoost	876	23	142	19

Individual confusion matrices for logistic regression, Random Forest, and XGBoost are shown in Figs. 6–8. A combined plot with all three matrices side by side is provided in Fig. 9. [7]–[11]

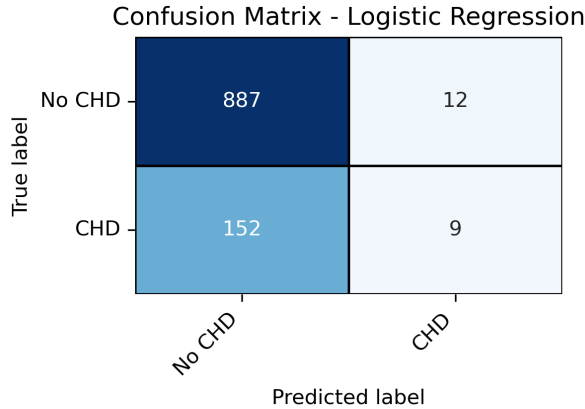


Fig. 6. Confusion matrix for logistic regression on the test set.

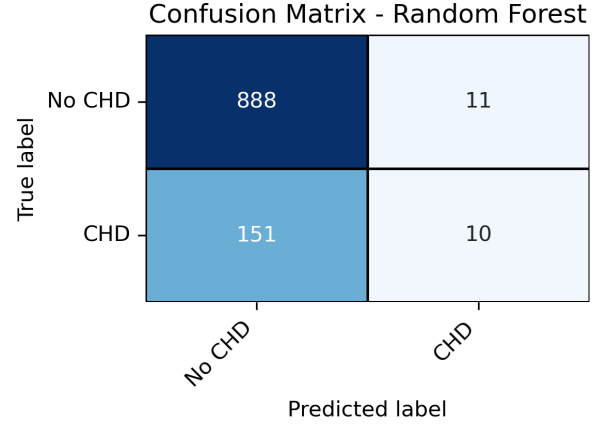


Fig. 7. Confusion matrix for Random Forest on the test set.

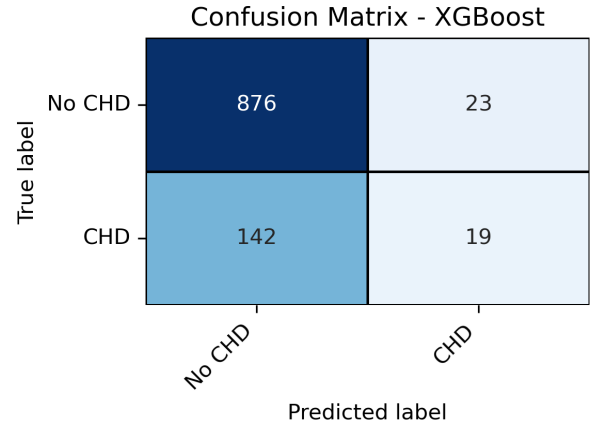


Fig. 8. Confusion matrix for XGBoost on the test set.

C. ROC Curves

ROC curves for all three models are plotted on the same axes in Fig. 10. The ROC-AUC values reflect moderate discriminative ability, with logistic regression achieving the highest AUC (0.7044), followed by XGBoost (0.6724) and Random Forest (0.6656). [7]–[11]

D. Feature Importance

Feature importance values from the Random Forest model highlight *sysBP*, BMI, age, *totChol*, *glucose*, *diaBP*, *heartRate*, and *cigsPerDay* as the most influential predictors, with *sysBP* and BMI having the highest importance scores. [7]–[9] The importance plot is shown in Fig. 11.

These quantitative results and visualisations summarise how each model performs in terms of discrimination and error patterns. In the following discussion, we interpret these findings in the context of screening for future CHD events.

IV. DISCUSSION

The results indicate that all three models achieve similar overall accuracy (around 84–85%), but they differ markedly in

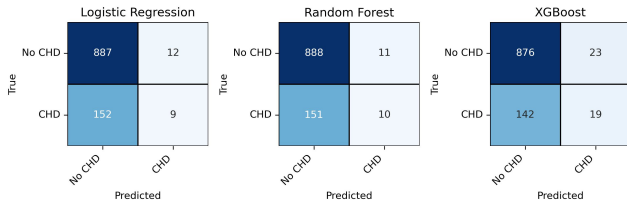


Fig. 9. Combined confusion matrices for logistic regression, Random Forest, and XGBoost on the test set.

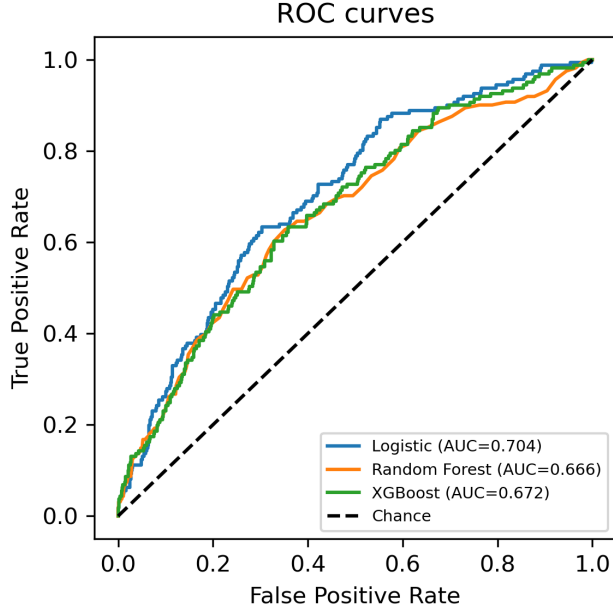


Fig. 10. ROC curves for logistic regression, Random Forest, and XGBoost on the test set.

how they handle the minority CHD class. Logistic regression and Random Forest yield the highest accuracy and AUC values but exhibit very low recall (5.6% and 6.2%, respectively), meaning that they miss most individuals who actually develop CHD. XGBoost, in contrast, achieves a slightly lower accuracy but more than doubles recall to 11.8% and yields the highest F1-score among the three models. [7]–[11]

In a screening context, where the goal is to flag high-risk individuals for further evaluation, a model like XGBoost that sacrifices some specificity to improve sensitivity may be preferable, provided that the increase in false positives is acceptable. The ROC curves and confusion matrices highlight this trade-off: XGBoost moves along the ROC curve toward higher true-positive rates at the cost of more false positives, while logistic regression and Random Forest operate near a more conservative threshold with fewer false positives but many false negatives. [12], [15] These findings align with previous work showing the importance of considering recall and F1-score, not just accuracy and AUC, when evaluating models on imbalanced medical datasets. [12]–[15]

These observations highlight that model choice and threshold selection must be aligned with clinical priorities rather than

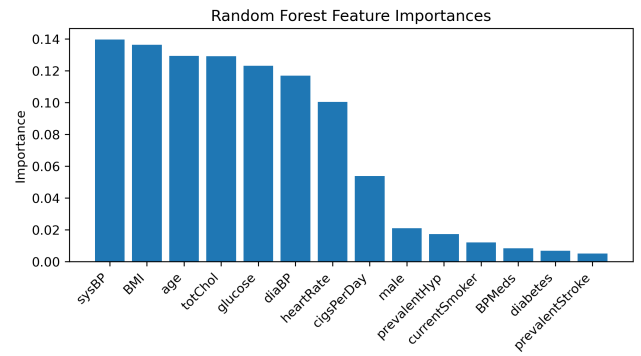


Fig. 11. Random Forest feature importance scores for the predictors used in the CHD risk model.

overall accuracy alone. The concluding section summarises the main insights and outlines directions for improving CHD risk prediction models.

V. CONCLUSION

This study set out to compare logistic regression, Random Forest, and XGBoost for predicting 10-year CHD risk using routinely collected variables from the Framingham Heart Study dataset, following a standardised preprocessing and evaluation pipeline. [1], [3], [5] All three models achieved similar overall accuracy (around 84–85%), but this metric alone was misleading because of the strong class imbalance between non-CHD and CHD cases. [2], [12], [15] Logistic regression and Random Forest obtained the highest accuracy and ROC–AUC, yet their recall for the positive CHD class remained extremely low, meaning that most future CHD events were still missed. [7]–[9] XGBoost, in contrast, delivered a more favourable trade-off by sacrificing a small amount of accuracy to more than double recall and achieve the highest F1-score, which is more appropriate for screening where missed high-risk patients are costly. [10], [11], [15]

From a practical standpoint, the results suggest that tree-based gradient boosting models such as XGBoost are better suited than standard logistic regression or untuned Random Forests when the clinical objective prioritises sensitivity over raw accuracy on imbalanced data. [7]–[11] However, the absolute performance of all three models in this single-centre dataset remains moderate, indicating that they should support, rather than replace, clinician judgement in CHD risk assessment. [12], [15] Future work will focus on systematic hyperparameter optimisation, explicit class-imbalance handling, and external validation on independent cohorts to improve robustness and generalisability of CHD risk prediction models. [7]–[11], [15]

ACKNOWLEDGMENT

The author would like to thank the faculty and staff of St Joseph’s University for their guidance and support.

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